WIL Project: Milestone 1

SkillGap: A Test-Driven RAG Career Navigator for International Data and IT Students

Group ID: WIL Project 84

Project Links

Trello Board: https://trello.com/b/1GEPr6pe/wil-project

Private GitHub Repository: https://github.com/Arulmozhi2003/WIL-project

Project Aim

To develop and evaluate a Test-Driven RAG assistant that helps international data and IT students understand Australian graduate job requirements and identify evidence-based skill gaps for their target roles.

User Story: As an international data or IT student, I want to understand the skills commonly required for my target graduate role so that I can identify and prioritise gaps in my current skill set.

Team Members

ID	Name	Proposed Role	Skills / Strengths	Contribution
s4138509	Arulmozhi Ezhilarasi Arunachalam	RAG Evaluation & Project Coordination	Python, Walert reproduction, retrieval evaluation, project coordination	20%
s4118043	Dharani Suresh	Data Collection & Knowledge Base Development	Data collection, preprocessing, source organisation, data quality	20%
s4184049	Lingkesh Natanasabapathy	Retrieval & RAG Development	Python, embeddings, retrieval, vector databases, RAG development	20%
s4135999	Prakash Sudarsanraj	Testing & Evaluation	Test design, quantitative evaluation, result comparison, performance analysis	20%
s4136157	Sakthisree Gaddam	Documentation, Ethics & User Experience	Technical writing, ethics, user requirements, presentation support	20%

Progress So Far

The group established shared Trello and private GitHub workspaces, reviewed the Walert project, discussed potential application domains, and selected a tentative project direction focused on helping international data and IT students understand Australian graduate job requirements.

As a preliminary reproduction of the Walert RAG approach, the quantitative evaluation environment was configured using Python 3.9. Dependency and Apple Silicon compatibility issues were identified and resolved before successfully executing Walert's repository-provided retrieval evaluation.

The evaluation compared the original intent-based system with BM25 RAG and dense/FAISS RAG. For **known topics**, dense/FAISS achieved the strongest performance with NDCG@1 = 0.6905, NDCG@3 = 0.6119 and NDCG@5 = 0.5812. For **inferred topics**, dense/FAISS achieved the highest NDCG@1 of 0.2500, while BM25 achieved the highest NDCG@3 of 0.2566 and NDCG@5 of 0.3291.

These preliminary results provide an initial quantitative baseline and demonstrate how alternative retrieval approaches can be systematically compared before developing the team's domain-specific Test-Driven RAG system.

Plan for the Next Three Weeks

Week	Planned Activities	Expected Output
1	Consolidate Walert reproduction findings; refine the stakeholder, project scope and user requirements; define initial evaluation criteria.	Confirmed project scope, user stories and evaluation framework.
2	Collect and clean a manageable Australian graduate-job knowledge base; develop the initial SkillGap RAG pipeline; prepare representative test questions and expected evidence.	Functional baseline RAG prototype, structured knowledge base and preliminary test set.
3	Evaluate the baseline using retrieval, correctness, faithfulness, source attribution and refusal-related measures; analyse failure cases and test one controlled improvement.	Quantitative baseline results, failure analysis and baseline-versus-improved comparison.

Preliminary Risks and Bottlenecks

Potential risks include rapidly changing job-advertisement data, difficulty defining reliable ground-truth answers, LLM cost or compute limitations, and excessive project scope. These risks will be managed using a fixed dataset snapshot, explicit evaluation criteria, a limited set of target roles, and preference for no-cost or locally executable tools where practical.